

Select efficient calculation strategies for unfamiliar problems

Choose and use the most efficient method to solve each problem. Show your thinking clearly.

Step 59

★ Today I am learning to select efficient calculation strategies for unfamiliar problems.

1. Whole Numbers

Use mental strategies, place value and rounding to estimate, then calculate.

a) $348 + 497 + 652$

 Estimate: _____
 Strategy: _____
 Answer: _____

b) $2,498 - 1,687$

 Estimate: _____
 Strategy: _____
 Answer: _____

c) $5,736 + 3,998 + 2,264$

 Estimate: _____
 Strategy: _____
 Answer: _____

d) $7,845 - 2,976 - 1,329$

 Estimate: _____
 Strategy: _____
 Answer: _____

e) 25×198

 Estimate: _____
 Strategy: _____
 Answer: _____

f) 125×64

 Estimate: _____
 Strategy: _____
 Answer: _____

g) $2,499 \times 34$

 Estimate: _____
 Strategy: _____
 Answer: _____

h) $48 \times 203 + 52 \times 203$

 Estimate: _____
 Strategy: _____
 Answer: _____

Think about it!

Which strategy might be best?

- Mental calculation
- Rounding and adjusting
- Compatible numbers
- Split or rearrange
- Use known facts
- Standard algorithm

Choose the easiest and most efficient method.

2. Fractions and Decimals

Choose efficient strategies to calculate each expression.

a) $\frac{3}{4} + \frac{2}{5} + \frac{1}{4}$

Strategy: _____ Answer: _____

b) $\frac{7}{8} - \frac{3}{4}$

Strategy: _____ Answer: _____

c) $2.5 + 3.75 + 1.25$

Strategy: _____ Answer: _____

d) $6.8 - 2.45$

Strategy: _____ Answer: _____

e) $\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4}$

Strategy: _____ Answer: _____

f) 2.5×1.6

Strategy: _____ Answer: _____

g) $4.8 \div 0.16$

Strategy: _____ Answer: _____

h) $7.2 \div 0.25$

Strategy: _____ Answer: _____

Strategies to consider:

- Find a common denominator
- Simplify before you calculate
- Use place value
- Multiply/divide by powers of 10
- Convert to easier equivalents

3. Mixed Operations

Decide the order and method that makes each calculation easiest.

a) $48 - 15 \times 6 + 27$

 Strategy: _____
 Work: _____
 Answer: _____

b) $144 \div 12 + 18 \times 5 - 42$

 Strategy: _____
 Work: _____
 Answer: _____

c) $(75 - 28) \times 6 + 36 \div 4$

 Strategy: _____
 Work: _____
 Answer: _____

d) $90 - (24 + 16 \div 4) \times 3$

 Strategy: _____
 Work: _____
 Answer: _____

Think about it!

Use BIDMAS/BODMAS. Calculate brackets first. Look for easy numbers or parts you can do mentally.

4. Algebraic Expressions

Simplify each expression in the most efficient way.

a) $4x + 7x - 5x$

Answer: _____

b) $3(a + 5) - 2(a - 4)$

Answer: _____

c) $2(x - 3) + 4(x + 1)$

Answer: _____

d) $(x + 6)^2$ (when $x = 4$)

Answer: _____

e) $5y - 2(3y - 1)$

Answer: _____

f) $(2x + 3)^2$ (when $x = 5$)

Answer: _____

g) $6a + 4b + 3a - b$

Answer: _____

h) $9(x - 2) - 3(x + 4)$

Answer: _____

Strategies to consider:

- Collect like terms
- Expand brackets
- Factorise (if helpful)
- Substitute values early
- Use the distributive law

5. Real-World Problems

Solve each problem. Choose the most efficient strategy.

a) A shop sells 4 packs of pens. Each pack has 25 pens. A school buys 18 packs. How many pens does it buy?

 Work: _____
 Answer: _____

b) A recipe uses 125 kg of flour for 1 batch. How much flour is needed for 6 batches?

 Strategy: _____
 Work: _____
 Answer: _____

c) A car travels 315 km in 5 hours. At the same speed, how far will it travel in 8.5 hours?

 Strategy: _____
 Work: _____
 Answer: _____

d) A computer is reduced from £860 to £688. What is the percentage reduction (to the nearest 1%)?

 Strategy: _____
 Work: _____
 Answer: _____

Think about it!

Read carefully. What is being asked? Estimate first. Then choose the best strategy to solve it.

6. Which Strategy?

Tick (✓) the strategy you think is most efficient. Then solve.

Question	Strategies to choose				Answer
	Mental	Rounding	Compatible Numbers	Standard Algorithm	
a) $199 + 298 + 401$	☐	☐	☐	☐	_____
b) $999 - 497$	☐	☐	☐	☐	_____
c) 25×48	☐	☐	☐	☐	_____
d) $\frac{3}{4} + \frac{5}{6}$	☐	☐	☐	☐	_____
e) 7.8×0.25	☐	☐	☐	☐	_____

Remember:

- There is not always one right answer.
- The best strategy saves time and reduces errors.
- Check your answer makes sense.

7. Challenge

Use your own efficient strategies to solve these problems.

a) $1,234 \times 37$

 Strategy: _____
 Work: _____
 Answer: _____

b) $(2.4 + 3.6) \times 25 - 15.5$

 Strategy: _____
 Work: _____
 Answer: _____

c) $\frac{15}{16} \div \frac{3}{8}$

 Strategy: _____
 Work: _____
 Answer: _____

d) $(x + 3)(x - 2)$ when $x = 7$

 Strategy: _____
 Work: _____
 Answer: _____

Challenge yourself!

Explain why your strategy was efficient.

My Learning Check

Circle how confident you feel.



I can do it!



Almost there!



I need more practice.



I need help.

Parent Observation (optional)

Please tick the boxes that apply.

- | | |
|--|--|
| ☐ Understands concepts | ☐ Explains thinking clearly |
| ☐ Uses strategies efficiently | ☐ Needs more practice |
| ☐ Solves problems confidently | |

Notes: _____

Today I learned...

1 thing I did well: _____

1 thing I will improve: _____